THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 9: **Some Applications of Trigonometry** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw a neat figure for every question of Sections B, C and D. Take $\sqrt{3} = 1.732$ where needed.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A 10 m long ladder makes an angle of 60° with the ground. The height it reaches on the wall is: 1

(a) $5\sqrt{3}$ m **(b)** 5 m **(c)** $10\sqrt{3}$ m **(d)** $\frac{10}{\sqrt{3}}$ m

2. When the sun's elevation is 45°, the shadow of a 12 m pole is: 1

(a) 12 m **(b)** 6 m **(c)** $12\sqrt{3}$ m **(d)** 24 m

3. The angle of elevation of the top of a 30 m tower from a point $30\sqrt{3}$ m from its foot is: 1

(a) 30° **(b)** 45° **(c)** 60° **(d)** 90°

4. When an observer looks at an object below the horizontal level, the angle between the line of sight and the horizontal is the: 1

(a) angle of elevation **(b)** angle of depression
(c) right angle **(d)** reflex angle

5. A student states: "As an observer walks towards a tower, the angle of elevation of its top decreases." The correct response is: 1

(a) Correct **(b)** Incorrect — it increases
(c) It remains the same **(d)** Cannot be determined

6. A kite is flying at a height of 60 m; its string makes 60° with the ground. The length of the string is: 1

(a) $40\sqrt{3}$ m **(b)** $60\sqrt{3}$ m **(c)** 120 m **(d)** $30\sqrt{3}$ m

7. An observer 1.5 m tall is 28.5 m from a chimney; the angle of elevation of its top from his eyes is 45°. Student A says the chimney is 28.5 m tall; Student B says 30 m. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

8. If the shadow of a tower is $\sqrt{3}$ times its height, the sun's altitude is: 1

(a) 30° **(b)** 45° **(c)** 60° **(d)** 90°

9. From the top of a 60 m cliff, the angle of depression of a boat is 60°. The distance of the boat from the foot of the cliff is: 1

(a) $20\sqrt{3}$ m **(b)** $60\sqrt{3}$ m **(c)** 30 m **(d)** 20 m

10. If the height of a tower equals the distance of the point of observation from its foot, the angle of elevation is: 1

(a) 30° **(b)** 45° **(c)** 60° **(d)** 90°

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A circus artist climbs a 20 m long rope tightly stretched and tied from the top of a vertical pole to the ground. Find the height of the pole if the rope makes 30° with the ground. 2

12. From a point P on the ground, the angle of elevation of the top of a 10 m tall building is 30°. Find the distance of P from the foot of the building. 2

13. A slide for children has its top at a height of 1.5 m and is inclined at 30° to the ground. Find the length of the slide. 2

14. A student says that the angle of elevation of the top of a 20 m tower from a point 20 m from its foot is 60°. Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. The shadow of a tower standing on level ground is found to be 40 m longer when the sun's altitude is 30° than when it is 60°. Find the height of the tower. 3

16. A 1.5 m tall boy is standing at some distance from a 30 m tall building. The angle of elevation from his eyes to the top of the building increases from 30° to 60° as he walks towards the building. Find the distance he walked. 3

17. Two poles of equal heights stand on either side of a road 80 m wide. At a point between them on the road, the angles of elevation of the tops of the poles are 60° and 30°. Find the height of the poles and the distances of the point from the poles. 3

OR

A TV tower stands vertically on a bank of a canal. From a point on the other bank directly opposite the tower, the angle of elevation of the top is 60°. From another point 20 m away from this point on the line joining it to the foot of the tower, the angle of elevation is 30°. Find the height of the tower and the width of the canal. 3

18. A 1.6 m tall statue stands on the top of a pedestal. From a point on the ground, the angle of elevation of the top of the statue is 60° and that of the top of the pedestal is 45°. Find the height of the pedestal. 3

OR

As observed from the top of a 75 m high lighthouse, the angles of depression of two ships are 30° and 45°. If one ship is exactly behind the other on the same side of the lighthouse, find the distance between the ships. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. The angle of elevation of a jet plane from a point A on the ground is 60°. After a flight of 30 seconds, the angle of elevation changes to 30°. If the jet is flying at a constant height of $3600\sqrt{3}$ m, find the speed of the jet in km/h. 5

OR

A straight highway leads to the foot of a tower. A man standing at the top of the tower observes a car at an angle of depression of 30°, approaching the foot of the tower with uniform speed. Six seconds later, the angle of depression of the car is 60°. Find the time taken by the car to reach the foot of the tower from this point. 5

20. **Case study:** Two men on either side of a cliff 80 m high observe the angles of elevation of the top of the cliff to be 30° and 60° respectively. 5

(i) Draw a labelled figure.

(ii) Find the distance of each man from the foot of the cliff.

(iii) Find the distance between the two men.

(iv) Which man is nearer the cliff and why?

OR

Case study: A window of a house is 10 m above the ground. From the window, the angles of elevation and depression of the top and bottom of another house on the opposite side of the lane are 60° and 45° respectively. 5

(i) Draw a figure.

(ii) Find the width of the lane.

(iii) Find the height of the other house.

END OF QUESTION PAPER — Total: 40 Marks